Program Outcomes and Course Outcomes

PROGRAM OUTCOME: B.A.

B.A. Programme enables the students to gain a thorough knowledge and skills in the disciplines of Arts stream. This will equip the students to analyse the past, present and plan for the future with a better understanding of the society.

Program Specific Outcome: History

I. Students shall be able to demonstrate specific understanding of History: Pre-history, Ancient History, Medieval History, Modern History, and Contemporary History along with the identification of the sources of History
II. Students shall be able to examine the History of religions, customs, institutions and administration
III. Develop thinking skills by analyzing, synthesizing, and evaluating historical sources. And develop an understanding of the social, political, religious and economic conditions of the people
IV. Analyze the relationship between the past and the present
V. Develop practical skills by identifying the boundaries and places on historical maps
VI. Enables the students to develop knowledge and competency required for research, profession, career and competitive examinations like IAS, KAS, ASI, etc.

Course Outcome:

Semester	Subject	Subject Code	Outcome
I	India in the Early Historical Period (to A.D.300)	BASHTC ï 102	¶ Students will gain knowledge about sources of history, historiography and geographical features ¶ Gain understanding about pre-historic period, Harppan culture, Vedic Age, Mahajanapadas, Mauryan Empire and Post-Mauryan India.
II	India in the Early Medieval Period (A.D. 300-1300)	BASHTC ï 152	¶ Understanding about the Age of the Guptas Chalukyas, Pallavas and Cholas ¶ Familiarising about the age of Rajputs and Muslim Invasion in India.
III	Medieval India (A.D. 1206-1556)	BASHTC ï 202	¶ Gaining knowledge about Delhi Sultanate, Economy, Society, and Polity ¶ Have better understanding about Vijayanagar Empire and South India in Early 14th Century A.D.

			¶ Information about Afghan Mughal Struggle and consolidation of the Mughal Empire
IV	Early Modern India (A.D. 1605-1856)	BASHTC ï 252	¶ Identify the condition of India under the Mughal domination ¶ Explain the Polity, Society, Culture, Literature, Music, art and architecture ¶ Analyse about the rise of the Marathas ¶ Examine the early phase of European Domination and consolidation of the British Empire
V	Colonial India (A.D. 1856-1885)	BASHTC ï 304	¶ Understand about Colonialism, Government under East India Company, Society and Culture ¶ Gain a thorough knowledge about the 1857 movement, genesis of Indian Nationalism and colonial policy in the post mutiny India
VI	History of Europe (A.D. 1789-1990)	BASHTC ï 305	¶ Understand causes and results of French Revolution, Rise of Napoleon, Unification of Italy and Germany ¶ Examine the World Wars I &II, League of Nations and UNO
	Making of the Indian Nation (A.D. 1885-1964)	BASHTC ï 355	¶ Understanding three phases of the Indian National Movement, Gandhi and Struggle for Swaraj ¶ Understanding the role of Shubhas Chandra Bose and INA ¶ Examine the social cultural aspects and legacy of freedom movement ¶ Understanding of Karnataka in 16th C, Karnataka after Vijayanagara, Karnataka towards colonial Domination and British rule
	History of Karnataka (A.D. 1565-1956)	BASHTC ï 356	¶ Understanding about rendition of Mysore, Social, Cultural and Political Developments in Mysore ¶ Understanding the unification of Karnataka

PROGRAM OUTCOME: B.A.

B.A. with Combinations: HEP & JKP

Political Science is studied as a part of History, Economics, and Political Science group, as well as with Journalism, Kannada, and Political Science group for B.A. degree. The study of Political Science enables the students to have better understanding of Politics, Political Theory, Political Systems, Constitution, Government, Governance, Public Administration, Human Rights and International Relations.

Program Specific Outcome: Political Science

The Political Science as a discipline enables the students to understand the basic principles of Politics and Governance including governing institutions and their bodies at various levels, political wings and organizations, including political behaviour. It helps the students to understand the working of various administrative bodies through Public Administration and Management. It also helps in understanding government and politics in a global context through Comparative Politics and International Relations. In totality, Political Science equips a student to pursue Civil Services Examinations and other Competitive Exams. Student will be able to pursue their career in teaching and research.

Course Outcome:

Semester	Subject	Subject Code	Outcome
I	Introduction to Political Science	BASPSC 102	¶ Understanding of the basic concepts and their divergent views ¶ Having a sensitive and broader vision of politics
II	Comparative Government and Politics	BASPSC 152	¶ Students gain the knowledge about the major Political Systems, operations on a comparative basis
III	Indian Political System	BASPSC 202	¶ Students gain awareness on Indian Political System ¶ Develop understanding and a sense of respect towards the Constitution of India ¶ Students gain awareness on Fundamental Rights and Duties
IV	Political Thinkers and Ideologies	BASPSC 252	¶ Students gain understanding about Western and Indian Political Thinkers and their political thoughts.
V	Public Administration	BASPSC 303	¶ Understanding the basics of public Administration ¶ Understanding the working of public institutions, civil services and organisational structure

			¶ Acquire knowledge about planning and budget making processes
	Introduction to International Relations	BASPSC 304	¶ Familiarity about nature, working, and need of International Relations ¶ Gain awareness about International Peace and Security
VI	Contemporary Issues and Trends in International Relations	BASPSC 353	¶ Gain knowledge about trends in International Relations and Global issues
	Theory and Practice of Management	BASPSC 354	¶ Development of Management skills and techniques ¶ Understanding of new developments in management

B.COM Program Outcome

1. The course focuses mainly on enhancing the employability skills of the commerce students.
2. It make a student employable and at the same time confident in his/her day to day transactions.
3. The course also meets the requirement of the young and enterprising Indians to nurture their dreams of entrepreneurship.
4. The course enables the students to pursue higher education in the respective stream.
5. The course helps the student to take up professional courses like CA/CS/ICWA/ICMA.

Program Specific Outcome

B.Com General	B.Com Vocational
1. The course focuses mainly on enhancing the employability skills of the commerce students. 2. It make a student employable and at the same time confident in his/her day to day transactions. 3. The course also meets the requirement of the young and enterprising Indians to nurture their dreams of entrepreneurship. 4. The course enables the students to pursue higher education in the respective stream.	1. The course focuses mainly on enhancing the employability skills of the commerce students. 2. Gives the knowledge of Direct Tax and Indirect Tax. 3. Gives practical knowledge of e-filing of returns. 4. The course enables the students to pursue higher education in the respective stream. 5. The course helps the student to take up professional courses like CA/CS/ICWA/ ICMA.

Course Outcome: B.COM (General)

Semester	Subject	Subject Code	Outcome
I	Language: Kannada		
	Language: Hindi		
	Language: Sanskrit		
I	English		
I	Economics		
I	Financial Accounting I	BCMCMC 106	- Record business activity in the ledger - Methodological approach to describe the activities of business - Learn accounting standards - Financial reporting process
I	Principals Of Management	BCMCMC 107	1. To study about organization functions and organization structure. 2. To study various Theories of Management. 3. To know about selection and recruitment process 4. To study about training and

			development procedure.
I	Business Statistics and Mathematics	BCMCMC108	Descriptive Statistics- measures of central tendency and measures of dispersion. Index numbers like- CPI, and formulation using various methods. Matrices and determinants. Matrix algebra and solving simultaneous equations using matrices.
II	Language: Kannada	BCMCMC	
	Language: Hindi		
	Language: Sanskrit		
II	English		
II	Economics		
II	Financial Accounting II	BCMCMC 156	- Dealing in bills - Preparation of financial statements - Reconciliation of errors in recording business transactions - Educate the students in hire purchase , instalment and consignment system
II	Modern Banking	BCMCMC 157	1. To have knowledge of banking structure in India. 2. To have knowledge about general and special functions of banking. 3. To have knowledge about Negotiable Instruments.
II	Business Statistics and Mathematics	BCMCMC158	Correlation and Regression Analysis. Time Series. Basics of banking arithmetic like ïSimple and Compound Interest, Discount etc.
III	Language: Kannada		
	Language: Hindi		
	Language: Sanskrit		
III	English		
III	Business Taxation I	BCMCMC 205	1. It helps the student to prepare Income statements. 2. It helps to file the returns 3. Compute the deductions and allowances Helps to assess transactions for the purpose of taxation.
III	Economics		
III	Financial Accounting III	BCMCMC 202	1. How to calculate sacrifice ratio, new profit sharing ratio and gain ratio. 2. Treatment of goodwill- accounting

			standard 3 and 10 3. How to calculate the amount paid to executors account in case of death of a partner.
III	Cost and management Accounting I	BCMCMC 203	1. Understanding the concept of cost and financial accounting. 2. Preparation of cost sheet 3. Knowing different methods of pricing the materials. 4. Calculation of remuneration plans.
IV	Language: Kannada		
	Language: Hindi		
	Language: Sanskrit		
IV	English		
IV	Business Taxation II	BCMCMC 255	1. To study income under various heads 2. To study how to compute depreciation
IV	Economics		
IV	Cost and Management Accounting II	BCMCMC 253	1. Knowing different overheads and distribution of overheads. 2. Computing machine hour and labour hour rate in factory. 3. Understanding integrated accounting systems. 4. Preparing non integrated accounts and reconciliation statement.
IV	Financial Accounting	BCMCMC 252	1. Preparation of branch account 2. Preparation of departmental accounts 3. SEBI guidelines regarding underwriting of shares.
V	Financial Accounting V	BCMCMC 304	1. Students can prepare companies Financial Statement of companies Act of 2013 2. Gives practical knowledge of preparation of statement of account at the time of liquidation. 3. Calculation of various accounts and statement of Holding and Subsidiary Company.
V	Cost and Management Accounting III	BCMCMC 305	1. To study the cost of each job and each batch. 2. To ascertain the value of abnormal loss, abnormal gain and normal loss. To find out the Notional profit, reserves and profit& loss under contract account

V	Marketing	BCMCMC 302	1. To have knowledge about consumer behaviour 2. To know about advertising and sales management 3. To study about National and International Standards
V	Business Taxation III	BCMCMC 307	
V	Business Law	BCMCMC 301	1. To have knowledge about basic knowledge about law of the country 2. They will get to know about various intricacies of mercantile law.
V	Financial Management	BCMCMC 303	- To educate about the basic pre-requisites needed to get started career in finance - Educates students about how to help clients managing finance - Understand specific responsibilities of financial professionals - Tracking investment activities - Knowledge of stock exchange of India
VI	Cost and Management Accounting IV	BCMCMC 355	1. How to prepare fund flow statement. 2. To know the accounting standard 3 in cash flow statement 3. To know how to prepare the different budgets.
VI	Auditing	BCMCMC 352	1. To study about Company Audit. 2. To study about audit of Computerized accounts 3. To know about Vouching and Verification
VI	Financial Management	BCMCMC 353	- Developing cash management strategies - Knowledge about investment in Mutual funds - Learn how to become decision makers for stock holders and financial organizations - How to evaluate and price different types of securities

VI	Business Taxation IV	BCMCMC 357	1. Understanding goods and services tax Act provisions, rules, and regulations. 2. Registration procedure under GST. 3. Computation of taxes under GST. 4. Understandings custom provisions and payment of taxes.
VI	Financial accounting VI	BCMCMC354	1. Understanding of NPA, rebate on bill discounted and final accounts of banking company 2. Maintenance and preparation of final accounts at the time of amalgamation, merger, external and internal reconstruction. 3. Calculation of ratio for external and internal purpose of accounting, taxation, banking.
VI	Cost And Management Accounting		1. How to prepare fund flow statement. 2. To know the accounting standard 3 in cash flow statement 3. To know how to prepare the different budgets.

B.COM (VOCATIONAL) COURSE OUTCOME			
Semester	Subject	Subject Code	Outcome
I	Language: Kannada		
	Language: Hindi		
	Language: Sanskrit		
I	English		
I	Financial Accounting I	BCMCMC	
I	Business Statistics and Mathematics	BCMCMC 108	Descriptive Statistics- measures of central tendency and measures of dispersion. Index numbers like- CPI, and formulation using various methods. Matrices and determinants. Matrix algebra and solving simultaneous equations using matrices.
I	Income Tax Law And Practice I	BCMTPV 151	1. learn about the fundamentals of income tax and the scope. 2. Learning important definitions under Income tax Act 3. Learn about deductions under chapter VI A 4. Computing the gross income and total income
I	Goods and Service Tax Law and Practice I	BCMTPV 152	1. Understanding basis concepts under GST. 2. Learning various definitions under GST. 3. Understanding GST council, administration and structure 4. Learning about levy and collection of GST.
II	Language: Kannada	BCMCMC	
	Language: Hindi		
	Language: Sanskrit		
II	English		
II	Financial Accounting II	BCMCMC	
II	Business Statistics and Mathematics	BCMCMC 158	Correlation and Regression Analysis. Time Series. Basics of banking arithmetic like ïSimple and Compound Interest, Discount etc.
II	Income Tax Law And Practice II	BCMTPV 251	1. Learning about the different heads of income with its components. 2. Calculations of allowances and perquisites 3. Computing income under the head salaries

II	Goods and Service Tax Law and Practice II	BCMTPV 252	1. Learning about computation of GST payable 2. Understanding the importance of time of supply of service send goods 3. Computing place of supply of service 4. Computing value of taxable goods 5. Knowing about E-Way Bill and its implications.
III	Language: Kannada		
	Language: Hindi		
	Language: Sanskrit		
III	English		
III	Financial Accounting III	BCMCMC 307	4. It helps the student to prepare Income statements. 5. It helps to file the returns 6. Compute the deductions and allowances 7. Helps to assess transactions for the purpose of taxation.
III	Cost and management Accounting I	BCMCMC 203	1. Understanding the concept of cost and financial accounting. 2. Preparation of cost sheet 3. Knowing different methods of pricing the materials. 4. Calculation of remuneration plans.
III	Income tax Law and practice III		
III	Goods and Service tax Law III		1. Understanding the concepts of input tax credit under GST 2. Computing Input Tax Credit under various circumstances 3. Allowing credits and conditions foe allowing credit under GST.
IV	Language: Kannada		
	Language: Hindi		
	Language: Sanskrit		
IV	English		
IV	Cost and Management Accounting II	BCMCMC 253	1. Knowing different overheads and distribution of overheads. 2. Computing machine hour and labour hour rate in factory. 3. Understanding integrated accounting systems. 4. Preparing non integrated accounts and reconciliation statement.
IV	Financial		

	Accounting IV		
IV	Income tax Law and practice III		
IV	Goods and Service tax Law III		1. Knowing and learning account and documents under GST 2. Computation of interest, tax, penalty, fines under various circumstances 3. Liability to tax under GST in special cases 4. Understanding Appeals and revisions procedures.
V	Financial Accounting V	BCMCMC 304	1. Students can prepare companies Financial Statement of companies Act of 2013 2. Gives practical knowledge of preparation of statement of account at the time of liquidation. 3. Calculation of various accounts and statement of Holding and Subsidiary Company.
V	Cost and Management Accounting III		
V	Business Law		
V	Financial Management I		
V	Entrepreneurship Development I		- To educate about the basic pre-requisites needed to get started career in finance - Educates students about how to help clients managing finance - Understand specific responsibilities of financial professionals - Tracking investment activities - Knowledge of stock exchange of India
V	Advance Income Tax		
VI	Financial accounting VI	BCMCMC354	1. Understanding of NPA, rebate on bill discounted and final accounts of banking company 2. Maintenance and preparation of final accounts at the time of amalgamation, merger, external and internal reconstruction. 3. Calalulation of ratio for external

			and internal purpose of accounting, taxation, banking.
VI	Cost and Management Accointing IV		
VI	Company Law		
VI	Financial Management II		
VI	Entrepreneurship development II		- Developing cash management strategies - Knowledge about investment in Mutual funds - Learn how to become decision makers for stock holders and financial organizations How to evaluate and price different types of securities
VI	Customs Duty		

Department of Physics
PROGRAM OUTCOME: BSc
1) To understand the basic concept, fundamental principles and the scientific theories related to various scientific phenomena and their relevancies in day to day life. 2)To acquire skills in handling Scientific instruments, planning and performing in laboratory experiments 3) Develop Scientific outlook not only with respects to science subjects but also in all aspects related to life. 4) Analyse the given Scientific data critically and systematically and the ability to draw the objective conclusion 5) Understand set of physical laws, describing the motion of bodies under the influence of system of forces. 6) Able to relate the structure of atom and subatomic particle, 7) Apply conceptual understanding of the physics to general real-world situations. 8) Basic Knowledge for their Higher studies.
Program Specific Outcome
Physics Chemistry Maths :Students can understand material science better with this combination Physics Maths Statistics: Students can gain more knowledge about mathematical physics and statistical physics. Physics Maths Computer Science: Students will gain basic working principle of digital electronics.

Course Outcome			
Semester	Subject	Subject Code	Outcome
I	General Physics I	PHC 103	Student is expected to 1) To learn the conservation laws of energy and linear and angular momentum and apply them to solve problem. 2) Become familiar with various thermodynamic process and work done in each of these processes. 3) To learn the fundamentals of thermodynamics and laws of thermodynamics. 4) Gain the knowledge of Low temperature and pressure.
I	Practical -I	PHC104	Student is able to understand the concept of 1) Surface tension of water 2) Acceleration due to gravity using spiral spring and bar pendulum 3) Youngôs modulus of 4) Terminal velocity and coefficient of viscosity of the given liquid, to find the density of the unknown liquid. 5) Verification of parallel axes theorem and MI of the table
II	General Physics II	PHC153	The student will be 1) Able to understand the basics properties of matter, how Youngô modulus defines 2) Able to grasp the basic ideas of special theory of relativity such as length contraction, time dilations and mass energy invariance 3) Able to understand the introduction part of Astrophysics. 4) Able to study the fundamentals of simple harmonic motions, damped and forced oscillations and grasps the significance of quality factor and damping coefficients. 5)Able to analyze the wave patterns using Fourier Theorem.
II	Practical -II	PHC154	Student is able to understand the concept of 1)Interfacial tension between two different Liquids 2)Linear and material densities of a wire using Sonometer. 3)Verification of perpendicular axes theorem and determination of rigidity modulus of the wire. 4)Radius of gyration and equivalent length.

III	Optics	PHC203	Student will be expected 1) To Understand the basic concepts of wave optics and an ability to compute basic quantities in optics. 2) To use the principles of wave motion and superposition to explain the physics of polarization, interference and diffraction. 3) To gain confidence in their ability to apply mathematical methods to understand electromagnetic problems to real-life situations. 4) Have gained elaborated knowledge about electrostatics and laws governing the charge distribution. 5) To solve a variety of problems related to F a r a d a y ô s l a w o f i n equations. 6) To understand the relevance of displacement current in the context of electromagnetic wave propagation. 7) To learn about black body, radiation pressure, solar constant, estimation of surface tension of Sun, LASER and its applications, holography.
III	Practical III	PHC204	Student is able to understand the concept of 1) Interference and diffraction experimentally 2) Black body radiation through Stefan-Boltzmann law 3) Moment of inertia of irregular body 4) Melting point of solid using thermocouple 5) Specific rotation of sugar solution using polarimeter. 6) Frequency of a tuning fork using Helmholtz resonator.

IV	Electricity & X-ray Crystallography	PHC 253	Student will be expected to 1) Study in depth the transient current response of LR , CR and LCR circuits , which is essential in designing as well as understanding the working of electronic circuits 2) Understand the basic methods of solving electrical dc network using network theorem 3) Learn the different types of filters. 4) study the Force acting on a moving charge and torque on a current loop in a magnetic field 5) Understand the electrical and magnetic measurements 6) Study the X-ray crystallography, Miller indices and Structure of NaCl and KCl 7) Learn the Superconductivity, Meissner effect and applications of Superconductivity.
IV	Practical IV	PHC 254	Student is able to understand the concept of 1) I n t e r f e r e n c e t h r o u 2) Charge sensitivity of a BG 3) Bandwidth, quality factor and self inductance using Series resonance circuit 4) Charging of a capacitor 5) Verification of Maximum power transfer theorem.

V	Modern Physics	PHC 307	Student will be expected 1) To become familiar with Blackbody radiation, Ultraviolet catastrophe, Photoelectric effect and Compton Effect and hence be aware how quantum theory emerged 2) Have gained a clear knowledge about wave properties of particles, De Broglie waves and its implications on the uncertainty principle. 3) Study the Bohr Atom model in detail and understand about atomic excitations 4) To be capable of analyzing and solving problems using oral and written reasoning skills based on the concepts of modern physics 5) Have grasped the idea of Wave Mechanics and gain the concept of eigen values, eigen functions and learn the basic postulates of quantum mechanics 6) To find solution t for many systems such as particle in a box, Hydrogen Atom and familiarize with different quantum numbers. 7) Become familiar with molecular spectroscopy and have gained basic ideas regarding microwave spectroscopy, infrared spectroscopy and Raman Spectroscopy.
V	Condensed Matter Physics	PHC 308	Student will be expected to 1) Understand basic concepts and mathematical methods of solid state physics. 2) Understand how statistics of the microscopic world can be used to explain the thermal features of the macroscopic world. 3) Learn Einsteinôs a specific heat of solid 4) Learn about Nanomaterials and its applications 5) Understand Hall effect and its applications 6) Study the concept of Fermi energy and Boltzmann tail 7) Learn about LEDs, Solar cells 8) Electrical conductivity of a metal 9) Transistor, amplifier, hybrid model of the transistor

V	Practical V	PHC 309	Student will be expected to 1)Study the characteristics of pn diode, Zener diode and LED 2)Study the transistor characteristics 3)Learn the measurement of self-inductance using Andersons Bridge 4)Understand the concept of energy gap of a Thermistor 5)Learn the band width, Quality factor using parallel LCR circuit
VI	Nuclear Physics	PHC 357	Students are able to 1)Relate the structure of atom and subatomic particle 2) Gain a clear picture of nuclear composition and various nuclear models. 3) Have a deep knowledge about Radio activity, nuclear Fission and Nuclear Fusion, the relevance of nuclear transformation. 4)Understand the working of nuclear detectors and counters, realize the importance of Cosmic rays and its effects on earth 5) Become familiar with nuclear particles and different particle accelerators.
VI	Electronics	PHC358	Student will be expected to 1)Have a basic knowledge of semiconductor physics 2)Acquire knowledge about how a semiconductor diode rectifies an input ac signal 3) Learn how to construct a transistor amplifier and how its gain varies with frequency 4) Know about various number systems and their applications, flip flops and counters 5)Familiar with electronic devices and functioning 6)Learn the principles of analog modulation and demodulation 7) Understand basic pulse, digital and advanced communication systems. 8) Understand the theory and applications of satellite communications. 9) Learn the basics of mobile communication 10) Study of Op-Amp parameters and design of inverting and non-inverting amplifier.
VI	Practical VI	PHC359	Student is able to understand the concept of 1)Capacitance of capacitor and mutual inductance of a coil using B.G 2) Construction of OR, AND and NOT gate using diode and transistor. 3) Study of Wein Bridge Oscillator 4)Construction of Full wave Bridge rectifier and Common Emitter Amplifier,

Program Specific Outcome: **Chemistry**

After the completion of the course student will be able to learn periodic trends, chemistry of main group, d and f block elements. They will be able to predict type of hybridization, structures of given compounds. They will understand different processes involved in manufacture of some industrially important compounds. They will learn basic principles of organic chemistry. They will be able to predict and write mechanisms of various addition, elimination, substitution and condensation reactions. They will be well versed with functional group chemistry, important named reactions, basic natural product chemistry. They will be able to derive many thermodynamic relations and use them in problem solving. They will learn various molecular spectroscopic methods and their applications. They will come to know about various analytical methods. They develop analytical skills from the practical course.

Course Outcomes: Chemistry

Semester	Subject	Subject Code	Outcome
I	Chemistry	BSCCHC 103	After completion of the course students will beable to, 1. Draw Born-Haber cycle for ionic crystals. 2. Draw molecular orbital diagrams of homo and hetero nuclear diatomic molecules. 3. Calculate Miller indices for different planes in a cubic crystal system. 4. Determine crystal structure and Avagadro number. 5. Understood nature of bonding in organic molecules. 6. Learned many named reactions and mechanisms and their evidences.
		CH:102	1. Find out melting point and boiling point of compounds. 2. Find out elements and functional group of organic compounds. 3. Analyze mono and bi-functional organic compounds.
II	Chemistry	BSCCHC 153	After completion of the course students will be able to, 1. Differentiate different states of matter based on their physical properties. 2. Identify resemblances in chemical properties of elements and their compounds within each main group. 3. Understand different processes involved in large scale production of various industrially important compounds. 4. Select a particular reagent for given chemical transformation. 5. Write mechanism for different electrophilic addition reaction to carbon carbon multiple bonds.
		CH:152	1. Prepare standard solutions. 2. Estimate chemical constituents by neutralization, iodometric, complexometric and redox titration.

			3. Equiped with microscale experiments.
III	Chemistry	BSCCHC 203	After completion of the course students will be able to, 1. Derive basic thermodynamic relations. 2. Do calculations for entropy change during various physical and chemical processes etc 3. Calculate magnetic mom block elements. 4. Classify different compounds into various types of acids and bases. 5. Use redox potential data for various applications. 6. Predict reactivities of carbonyl compounds and mechanisms.
		CH: 202	1. Identify acid and basic radicals in a given inorganic salt mixture. 2. Develop qualitative analytical skills.
IV	Chemistry	BSCCHC 253	After completion of the course students will be able to, 1. Predict type of hybridization, geometry and magnetic properties of coordination compounds. 2. Derive thermodynamic relations for different chemical equalibria. 3. Write phase diagram for some one and two component systems. 4. Solve problems based on colligative properties. 5. Differentiate between S_N1 and S_N2 and E_1 and E_2 mechanism. 6. Write mechanisms for aromatic electrophilic and nucleophilic substitution.
		CH: 252	1. Determine physical parameters like viscosity, surface tension, refractive index etc. 2. Determine the rate constant and order of the reactions.
V	Chemistry	BSCCHC 307	After completion of the course students will be able to, 1. Understand uses of metal complexation in metallurgy and chemical analysis. 2. Predict reactivities of metal complexes based on thermodynamic and kinetic aspects. 3. Understand terms like transport number, equivalent conducatance etc. 4. Understand applications of conductance and EMF measurements. 5. Understand physical aspects of rotational vibrational Spectroscopy.

			6. Give E Z and R S nomenclature.
		BSCCHC 308	After completion of the course students will be able to, 1. Solve problems based on particles in one dimensional box. 2. Write orgel diagrams. 3. Understand applications of flame photometry thermoanalytical methods. 4. Understand roles of different metal ions in biological processes. 5. Understand applications of organometallic compounds as catalysts. 6. Predict structure and reactivities of basic heterocyclic compounds.
			1. Estimate the given compound quantitatively. 2. Determine adulterants in food stuffs. 3. Equip with quantitative analytical skills.
VI	Chemistry	BSCCHC 357	After completion of the course students will be able to, 1. Understand preparation and properties of inorganic and synthetic polymers. 2. Calculate quantum yield of photochemical reactions 3. Write various reactions of monosaccharides 4. Classify and predict stereochemistry of amino acids. 5. Predict acidity orders of various aliphatic and aromatic carboxylic acids 6. Write various reactions of carboxylic acids with mechanism.
		BSCCHC 368	After completion of the course students will be able to, 1. Differentiate between colorimetry and spectrophotometry 2. Interpret PMR spectra of simple organic molecules. 3. Understand principle of photoelectron spectroscopy 4. Understand principle and working of mass spectrometer. 5. Understand synthesis and mode of action of some drugs and chemotherapeutic agents. 6. Understand preparation, structure and properties of some pesticides and fungicides.
		CH:353	1. Develop instruments handling techniques. 2. Develop organic compound preparation skills. 3. Develop coordination complex preparation skills.